

PayNear

MERN Full-Stack Project Proposal

Prepared July 2026

1. Project Summary

PayNear is a location-based web application that helps users discover nearby establishments based on the payment methods they accept. It combines location awareness, search filters, verified establishment data, and user preferences so people can quickly choose where to go without worrying whether their preferred payment option is available.

2. Problem Statement

Consumers often discover only at checkout that an establishment does not accept their preferred payment method. Existing map and directory applications may show nearby places, but accepted payment options can be missing, outdated, or difficult to verify.

3. Proposed Solution

PayNear will provide a searchable directory of verified establishments and their accepted payment methods. With the user's permission, the application will use the user's current location, apply radius and payment-method filters, and rank suitable establishments by relevance and distance.

4. Project Objectives

- Help users find nearby establishments that accept a selected payment method.
- Provide clear, verified, and maintainable establishment information.
- Reduce uncertainty by combining location, availability, ratings, and user preferences.
- Provide administrators with controlled tools for verification and establishment data management.

5. Target Users

- Everyday consumers who use cash, cards, e-wallets, bank transfers, or other payment options.
- Travelers and commuters who need compatible establishments in an unfamiliar area.
- Administrators who verify listings and maintain establishment information.

6. MVP Features

High Priority

- Authentication and Registration: Users can register, sign in, sign out, and maintain a secure session.
- Location Detection with Consent: The application asks permission before detecting the user's location and provides a manual fallback.
- Nearby Search and Filters: Users can search within a target radius and filter establishments by payment method.
- Relevant Result Suggestions: Results are ranked using proximity and payment-method compatibility.
- Payment Availability Details: Each establishment shows its currently accepted payment methods.
- Admin Verification and CRUD: Administrators can verify, add, update, and deactivate establishment records.

Medium Priority

- Reviews and Ratings: Authenticated users can review establishments they visited.
- Favorites: Users can save or remove favorite establishments.
- Preferred Payment Method: Users can store a preferred payment method for faster searches.
- Shareable Establishment Link: Users can share a public link to establishment details.

7. Out of Scope for the Initial MVP

- Payment processing or storage of card and e-wallet credentials.
- Merchant self-registration and automated document verification.
- Turn-by-turn navigation and live traffic routing.
- Native Android or iOS applications.

8. High-Priority User Stories and Acceptance Criteria

US-01: Account Access (High Priority)

User Story: As a new or returning user, I want to register and sign in so that I can use personalized features securely.

Acceptance Criteria:

- Registration validates required fields and prevents duplicate email addresses.
- Valid credentials create an authenticated session, while invalid credentials return a clear error.
- A signed-in user can sign out, and protected routes reject unauthenticated access.

US-02: Location Permission (High Priority)

User Story: As a user, I want to control access to my location so that nearby results are useful without sacrificing consent.

Acceptance Criteria:

- The browser asks for permission before location detection begins.
- Approved access provides usable coordinates for the current search.
- Denied or unavailable access displays a manual location option and a helpful message.

US-03: Search and Filter (High Priority)

User Story: As a user, I want to filter nearby establishments by payment method and radius so that I only see relevant options.

Acceptance Criteria:

- A search requires a location and supports at least one payment-method filter.
- Returned establishments fall within the selected radius and match the selected payment method.
- Results provide clear loading, empty, and error states.

8. High-Priority User Stories (Continued)

US-04: Best-Match Results (High Priority)

User Story: As a user, I want suitable establishments ranked for me so that I can choose quickly.

Acceptance Criteria:

- Each result shows the name, distance, address, and accepted payment methods.
- Payment compatibility is based on the current establishment record.
- Default ordering prioritizes matching establishments by distance.

US-05: Establishment Details (High Priority)

User Story: As a user, I want to view verified payment information so that I know whether an establishment meets my needs.

Acceptance Criteria:

- The detail page lists accepted payment methods and the current listing status.
- The page identifies verified listings and provides address and location information.
- Deactivated establishments do not appear as valid search choices.

US-06: Admin Verification and Data Management (High Priority)

User Story: As an administrator, I want to verify and manage establishment records so that directory information remains reliable.

Acceptance Criteria:

- Only authorized administrators can access verification and CRUD actions.
- The administrator can create, update, verify, and deactivate an establishment.
- Required fields and verification evidence are validated, and saved changes update user-facing results.

9. Supporting User Stories and Acceptance Criteria

US-07: Preferred Payment Method (Medium Priority)

User Story: As a user, I want to save my preferred payment method so that searches begin with a relevant default.

Acceptance Criteria:

- The user can select a supported payment method during registration or in account settings.
- The saved preference is preselected during search and can be changed.

US-08: Favorites (Medium Priority)

User Story: As a user, I want to save or remove favorite establishments so that I can revisit them easily.

Acceptance Criteria:

- An authenticated user can add and remove an establishment from favorites.
- The favorites list persists across sessions and does not contain duplicates.

US-09: Reviews (Medium Priority)

User Story: As a user, I want to review an establishment I visited so that I can help others make informed choices.

Acceptance Criteria:

- An authenticated user can submit a rating and written review.
- The user can edit or delete their own review but cannot modify another user's review.
- The establishment page displays reviews and an updated aggregate rating.

US-10: Share a Place (Medium Priority)

User Story: As a user, I want to share an establishment link so that another person can open its location and payment details.

Acceptance Criteria:

- The share action produces a stable link to the establishment detail page.
- A recipient can open public details without being required to sign in.
- The shared view does not expose private account or precise user-location data.

10. Definition of MVP Complete

- All high-priority acceptance criteria pass functional testing.
- The main user journey works on mobile and desktop layouts.
- Role-based access protects administrative actions.
- The deployed application connects securely to its production database.

11. Technical Approach

PayNear will use a REST-based MERN architecture with geospatial search support and role-based access control.

Frontend

- React for the user interface and reusable components.
- React Router for page navigation and protected routes.
- Responsive forms, account pages, search results, and establishment details.

Backend

- Node.js and Express for REST endpoints, validation, authorization, and business rules.
- JWT and bcrypt for authentication, password hashing, and protected access.
- Role-based access control for administrator-only actions.

Database

- MongoDB and Mongoose for application data and schema validation.
- Geospatial indexes for nearby establishment queries.
- Main data entities: User, Establishment, Payment Method, Establishment Payment, and Review.

Location

- Browser Geolocation API for consent-based location detection.
- A compatible mapping service for maps and location display.

12. Quality and Security Requirements

- Responsive and accessible interface with loading, empty, success, and error feedback.
- Server-side validation, sanitized inputs, hashed passwords, and least-privilege authorization.
- Location data is used only with permission and is not exposed in shared links.
- Geospatial queries are indexed, and long result sets are paginated.

13. Development Plan

Phase 1 - Foundation

- Set up the MERN project, environment configuration, database schemas, and API conventions.

Phase 2 - Core Access

- Build registration, sign-in, session handling, protected routes, and administrator authorization.

Phase 3 - Discovery

- Build location consent, geospatial search, payment filters, result ranking, and establishment details.

Phase 4 - Administration

- Build admin CRUD, verification status, and payment availability management.

Phase 5 - Supporting Features

- Build preferences, favorites, reviews, ratings, and shareable links.

Phase 6 - Release

- Complete responsive testing, security checks, defect fixes, seed data, and deployment.

14. Success Measures

- Search results match the chosen payment method and fall within the requested radius.
- Only active and verified listings appear as trusted search results.
- A user can move from location permission to a suitable establishment through a clear flow.
- Authorized administrators can maintain data without direct database access.

15. Proposal Note

This document is intended for initial project approval. Wireframes and a Trello board are not included at this proposal stage and may be prepared after the project concept, MVP scope, and technical direction are approved.

16. Requested Decision

Approve PayNear as a MERN full-stack project and approve the proposed MVP scope as the basis for detailed design and implementation.

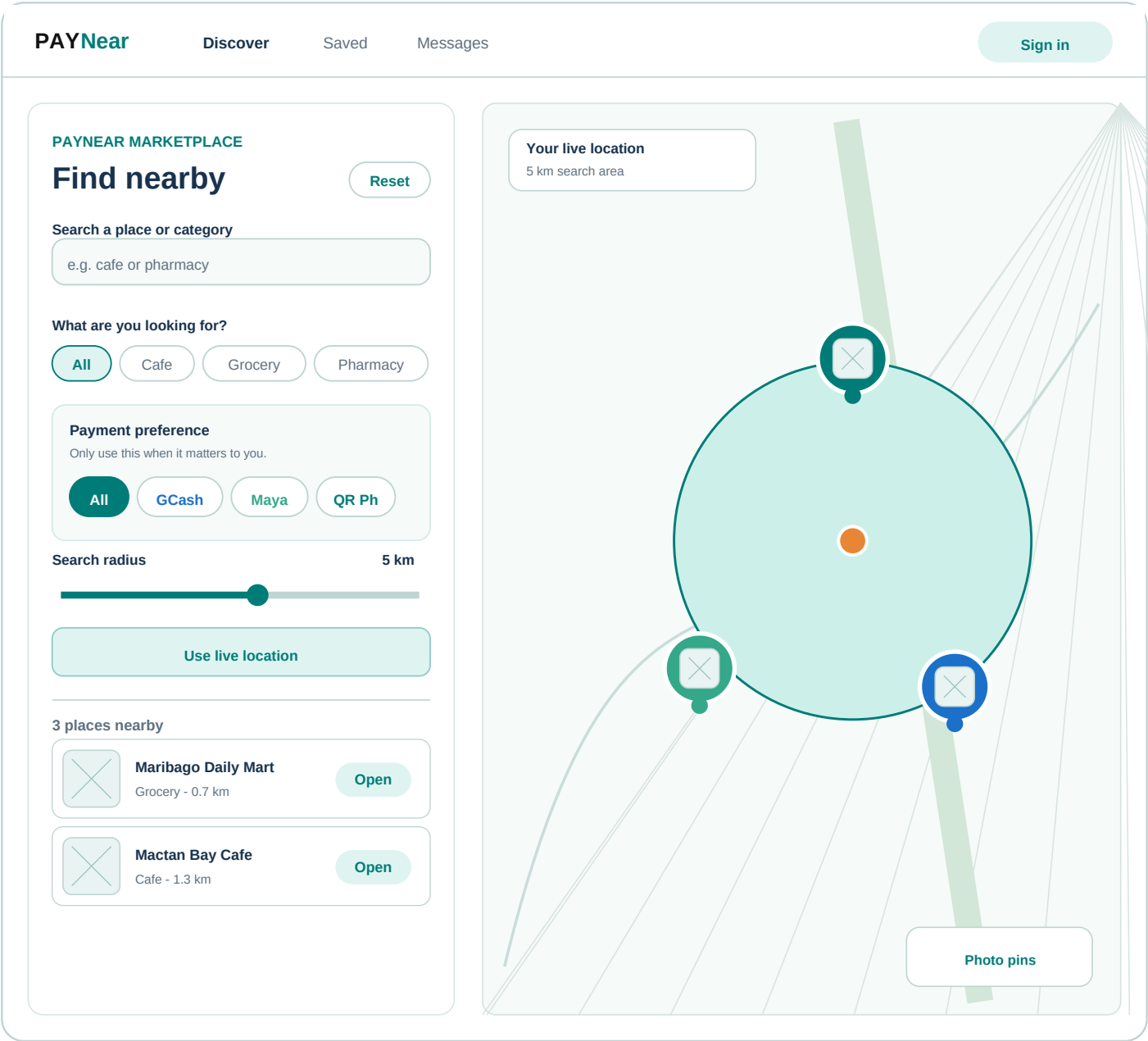
15. Updated Wireframes

Updated July 30, 2026. This section replaces the wireframe note on page 7.

WF-01

Discover nearby - primary desktop screen

One sidebar keeps search, payment availability, radius, and matching establishments in a single flow.



Wireframe notes

The map uses the visitor location after permission, with a visible radius circle. Store images identify establishments on the map; payment badges communicate accepted methods only and do not process a payment.

15. Updated Wireframes (continued)

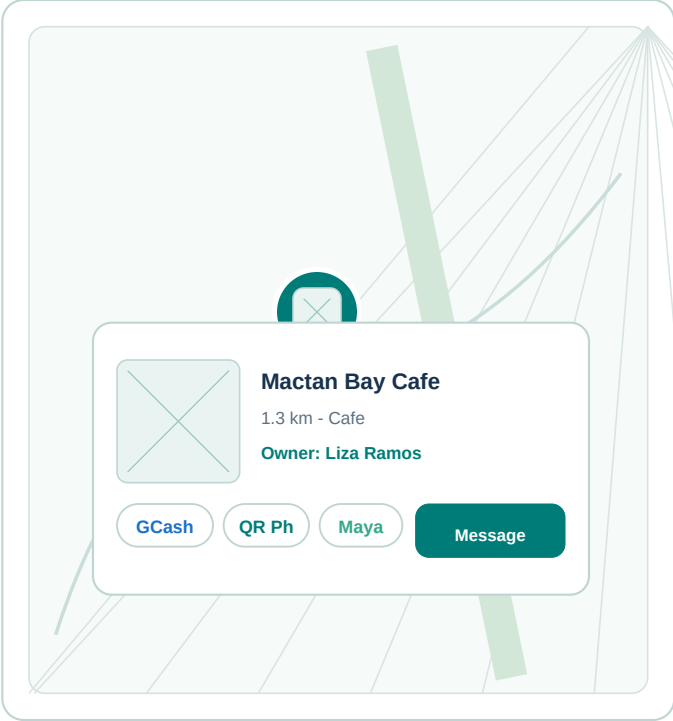
The visitor can inspect a photo pin, then begin a real-time conversation with the listed contact.

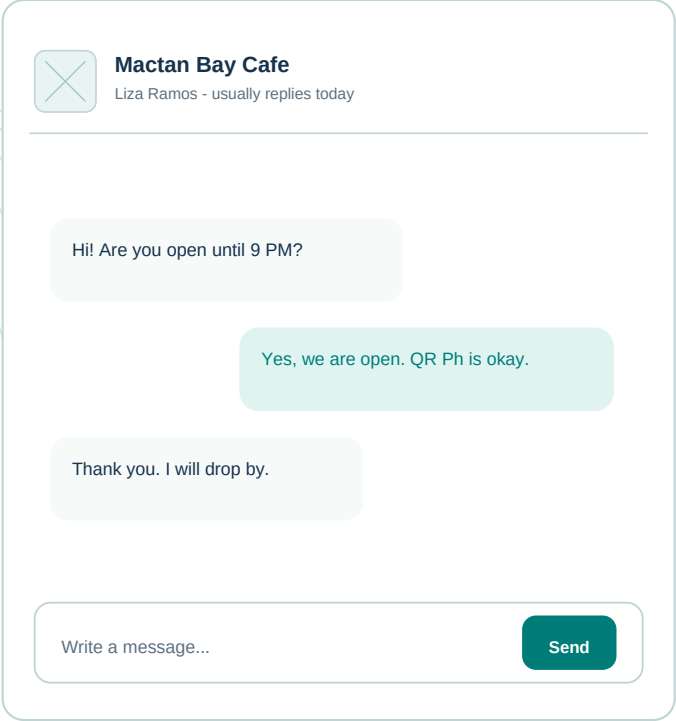
WF-02

Photo pin and establishment preview

WF-03

Real-time chat with owner





Tap a photo pin to view the listing, its owner, distance, accepted methods, and contact action.

Socket.IO keeps the conversation current for both visitor and establishment contact without reloads.

Interaction sequence



The preview does not imply payment processing. It simply helps visitors find an establishment whose listed payment options match their needs, then communicate directly with the establishment.

15. Updated Wireframes (continued)

THE LAST ROOM

Supporting screens keep listing data manageable for administrators and practical on a smaller mobile screen.

WF-04

Admin listing management

PayNear Admin

New listing

Establishment details

Name: Maribago Daily Mart

Category: Grocery

Address: Maribago, Lapu-Lapu City

Contact: Miguel Reyes

Listing photo

UPLOAD

Accepted payment methods

GCash

Maya

QR Ph

InstaPay

Save changes

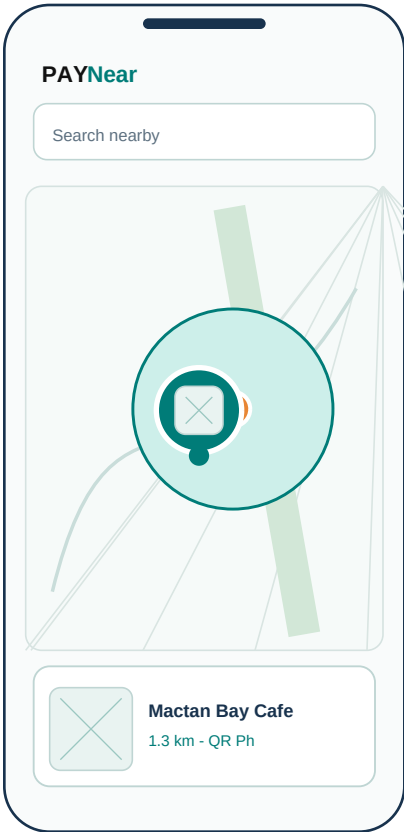
Archive

Verify

Admin tools manage listing details, images, availability, and verification.

WF-05

Mobile discovery



On mobile, search and the nearby result stay within reach while the radius remains visible.

Implementation alignment

Images

Listing photos are uploaded and become map identifiers.

Payments

GCash, Maya, QR Ph, InstaPay, banks, cards, cash, and transfer can be listed.

Roles

Visitors discover and message; administrators create, update, archive, and verify listings.

End of updated wireframes